

Day 5: Network Testing Tools - Answers

Tools: ping, traceroute (tracert), nslookup

Note: Network tests aapke apne computer par chalte hain, isliye neeche diye gaye output **sample (illustrative)** hain. Commands khud run karke apne asli numbers/IPs replace kar lena.

1. Ping - www.youtube.com

```
ping www.youtube.com (Windows: 4 packets; Linux/Mac: ping -c 4 www.youtube.com)
Reply from 142.250.195.78: bytes=32 time=21ms TTL=117
Reply from 142.250.195.78: bytes=32 time=19ms TTL=117
Reply from 142.250.195.78: bytes=32 time=23ms TTL=117
Reply from 142.250.195.78: bytes=32 time=21ms TTL=117
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
Approximate round trip times: Minimum = 19ms, Maximum = 23ms, Average = 21ms
```

Answer: Average response time = **21 ms** (sample), 0% packet loss. Matlab connection achha hai. Average value output ki last line mein milti hai.

2. Traceroute - www.flipkart.com

```
tracert www.flipkart.com (Linux/Mac: traceroute www.flipkart.com)
 1  2 ms  192.168.1.1      <- Home router / gateway
 2  8 ms  10.10.0.1             <- ISP local node
 3 12 ms  103.xx.xx.1           <- ISP backbone router
 4 15 ms  182.xx.xx.5           <- ISP core / peering router
 5 18 ms  72.14.xx.xx           <- Transit / cloud provider edge
 6 20 ms  www.flipkart.com [IP]  <- Destination
```

Answer: Intermediate hops (sample) = hop 1 se 5: home router, ISP ke local, backbone aur core routers, aur transit/edge router. Hop 6 destination server hai. Har hop ek network device (router) hai jisse aapka data guzarta hai. Agar kisi line mein ***** Request timed out** aaye to wo router ICMP reply block kar raha hai, zaroori nahi ki wo kharab ho.

3. nslookup - www.zomato.com

```
nslookup www.zomato.com
Server:      dns.google
Address:     8.8.8.8

Non-authoritative answer:
Name:        www.zomato.com
Addresses:   104.xx.xx.xx
             104.xx.xx.yy
```

Answer: Zomato ka IP address (sample) = **104.xx.xx.xx**. Aap apna asli IP yahan likhein. Ek domain ke multiple IPs aana normal hai (CDN/load balancing).

4. Local device ko ping karna

```
ping 192.168.1.15 (friend ke laptop / phone ka local IP)
```

Device online ho: Reply aata hai, time bahut kam (1-5 ms), 0% loss.

Device offline ho: *Request timed out* ya *Destination host unreachable* aata hai, aur 100% packet loss dikhta hai. Iska matlab device band hai, Wi-Fi se disconnected hai, ya firewall ICMP block kar raha hai (phone mein aksar power-saving ya firewall ki wajah se bhi hota hai).

5. Short Report (Tool selection)

Maine ping, traceroute aur nslookup teeno se test kiya. **(a) Website not loading:** sabse pehle **ping** useful hai, kyunki YouTube par 0% loss aur 21 ms average mila, jisse pata chalta hai ki internet chal raha hai; agar ping fail ho to connectivity problem hai. **(b) Slow connection:** **traceroute** best hai, kyunki ye dikhati hai ki Flipkart tak 6 hops mein se kis hop par delay achanak badh raha hai (jaise 20

ms se 200 ms), yani bottleneck kahan hai. **(c) DNS issues: nslookup** best hai, kyunki Zomato ke liye ye domain ko IP (104.xx.xx.xx) mein convert karta hai; agar ye fail ho jabki IP se ping chalta ho, to problem DNS ki hai. Is tarah teeno tools alag-alag layer ki problem pakadte hain: ping = reachability, traceroute = path/delay, nslookup = name resolution.

Problem	Best tool	Example (apne test se)
Website not loading	ping	youtube.com: 0% loss, avg 21 ms
Slow connection	traceroute	flipkart.com: kaun sa hop slow hai
DNS issue	nslookup	zomato.com -> 104.xx.xx.xx